TEXORA · LECTURE SHEET

Standard Costing, Joint and By-products, Direct Costing

Predetermined costs as a control instrument (the standard cost card), the apportionment of joint costs at the split-off point and the treatment of by-products, and the logic of direct (marginal) costing against absorption costing.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 5 · LEVEL 4 TERM 2

1. Standard costing

A **standard cost** is a predetermined cost — what a unit *should* cost under efficient, normal operating conditions — set in advance from engineering studies, time study and price forecasts. **Standard costing** values output at standard, compares with actual, and explains the difference as **variances** (Topic 9 computes them). Its uses: control by exception (only variances are investigated), realistic pricing and tendering before costs are incurred, performance evaluation, and simpler bookkeeping (stock moves at standard). Standards come in grades — ideal (perfection, usually demotivating), basic (frozen for trend), and **currently attainable** (efficient but reachable — the grade used in practice).

WORKED EXAMPLE — STANDARD COST CARD OF A SHIRT

Direct material: 3 kg-equivalent units of fabric & trims @ Tk 120 = Tk 360

Direct labour: 2 standard hours @ Tk 90 = Tk 180

Variable factory overhead: 2 hours @ Tk 25 = Tk 50

Fixed factory overhead: 2 hours @ Tk 40 = Tk 80

Standard cost per shirt = 360 + 180 + 50 + 80 = Tk 670.

Every element is a quantity × price pair, because variances split exactly that way: price/rate variances from the rate column, quantity/efficiency variances from the usage column (Topic 9 works both).

2. Joint products and by-products

A single process often yields several products. **Joint products** are two or more main products produced simultaneously from one process and one input (different grades of yarn or fabric qualities from one line; grades of recovered fibre). **By-products** are incidental outputs of minor value arising with the main product (cotton waste from spinning, selvedge cuttings, recovered dye-plant by-streams). Up to

the **split-off point** all costs are joint and inseparable; after it, each product bears its own further (separable) costs.

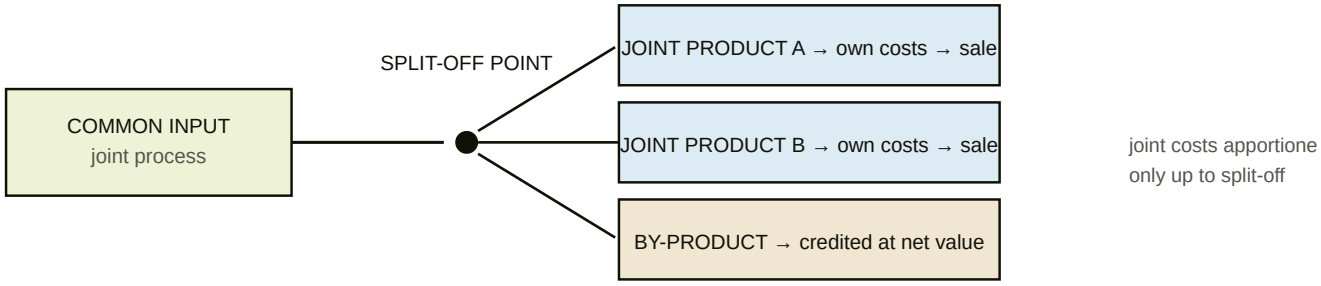


Figure 1 — The split-off point. Before it, one joint cost pool serves all products; after it, costs are separable and belong to each product alone.

WORKED EXAMPLE — APPORTIONING A JOINT COST, AND CREDITING A BY-PRODUCT

A joint process costs Tk 9,00,000 up to split-off and yields products A, B and C whose sales values at split-off are Tk 8,00,000, Tk 4,00,000 and Tk 3,00,000 (total Tk 15,00,000).
Apportioning the joint cost in the sales-value ratio:
A: 9,00,000 × 8/15 = **Tk 4,80,000** · B: 9,00,000 × 4/15 = **Tk 2,40,000** · C: 9,00,000 × 3/15 = **Tk 1,80,000** (total Tk 9,00,000 ✓).
(Other accepted bases: physical units, and net realisable value where further processing occurs. The sales-value basis charges most cost to the product best able to bear it — an allocation convention, not a physical truth.)

By-product: a main product's process costs Tk 9,60,000 for 9,000 units, and its by-product realises a net Tk 60,000 (sales minus its own disposal costs). The by-product value is credited to the process:
Cost per unit of main product = (9,60,000 − 60,000) ÷ 9,000 = **Tk 100 per unit**.
A by-product is never apportioned joint cost as if it were a joint product; it simply reduces the main product's cost.

3. Direct (marginal) costing versus absorption costing

Direct costing — in the British terminology, **marginal costing** — charges products only with variable (direct) production costs and treats all fixed production overhead as a **period cost** written off in the month it arises. **Absorption costing** charges products with variable *and* fixed production overhead (Topic 3's rates), so stock carries a share of fixed cost. The choice changes reported profit whenever production and sales differ, because under absorption some fixed cost hides in closing stock:

WORKED EXAMPLE — WHY THE TWO PROFITS DIFFER

A plant produces 10,000 units and sells 9,000. Fixed production overhead is Tk 90 per unit produced at this level. Closing stock = 1,000 units.

Under absorption costing, closing stock carries 1,000 × Tk 90 = **Tk 90,000 of fixed overhead into next period**; under marginal costing that Tk 90,000 is charged against this period's profit in full.

Result: absorption profit exceeds marginal profit by exactly **Tk 90,000** this period — reversed next period when the stock sells.

Marginal costing's own logic gives **break-even analysis** (Topic 6): with price Tk 500, variable cost Tk 320 and fixed costs Tk 9,00,000, contribution per unit = Tk 180 and break-even = 9,00,000 ÷ 180 = **5,000 units**.

Reconciliation rule: absorption profit − marginal profit = change in stock (units) × fixed overhead rate per unit.

ASPECT	MARGINAL (DIRECT) COSTING	ABSORPTION COSTING
Fixed production overhead	Period cost	Product cost, absorbed into units
Stock valuation	Variable cost only	Full production cost
Profit behaviour	Follows sales volume	Follows sales and production volumes
Best use	Decisions: CVP, pricing, make-or-buy	External reporting, full-cost pricing

Glossary

Standard cost — predetermined unit cost under normal efficiency. **Standard cost card** — the element-wise build-up of a standard cost. **Variance** — difference between standard and actual (computed in Topic 9). **Joint products** — main products emerging together from one process. **By-product** — incidental minor output, credited at net realisable value. **Split-off point** — where joint products become separately identifiable. **Marginal/direct costing** — variable-cost-only product costing. **Absorption costing** — full production-cost product costing.

Key points

• The standard cost card pairs a quantity with a price for every element (shirt: Tk 670) — that pairing is what makes variance analysis possible.

• Joint costs are apportioned up to split-off, commonly on sales value (Tk 9,00,000 → 4,80,000 / 2,40,000 / 1,80,000); by-products are simply credited at net value (→ Tk 100/unit).

• Marginal and absorption profits differ only through stock: 1,000 units × Tk 90 fixed = Tk 90,000 of difference, reversing when stock sells.

Practice

1. Apportion a joint cost of Tk 6,00,000 among products X, Y, Z whose sales values at split-off are Tk 5,00,000, Tk 3,00,000 and Tk 2,00,000.
2. Production 12,000 units, sales 11,500 units, fixed production overhead rate Tk 60 per unit. By how much, and in which direction, do absorption and marginal profits differ?
3. Build a standard cost card: material 2.5 kg @ Tk 140; labour 1.5 h @ Tk 110; variable overhead 1.5 h @ Tk 30; fixed overhead 1.5 h @ Tk 45.
4. Why is a by-product credited to the process rather than apportioned a share of joint cost "for fairness"?

Answer key

1. Total sales value = Tk 10,00,000. X: 6,00,000 × 5/10 = **Tk 3,00,000**; Y: 6,00,000 × 3/10 = **Tk 1,80,000**; Z: 6,00,000 × 2/10 = **Tk 1,20,000**. (Total Tk 6,00,000 ✓.)
2. Stock rises by 500 units × Tk 60 = **Tk 30,000**; absorption profit is **higher** by Tk 30,000, because that much fixed overhead sits in closing stock instead of the period's charges.
3. Material = 2.5 × 140 = Tk 350; labour = 1.5 × 110 = Tk 165; variable OH = 1.5 × 30 = Tk 45; fixed OH = 1.5 × 45 = Tk 67.50. **Standard cost = Tk 627.50 per unit**.
4. Joint-cost apportionment exists to value the stocks of main products whose profitability is being judged. A by-product is incidental and minor; giving it a share of joint cost would inflate an unimportant number, understate the main product's cost, and add bookkeeping out of all proportion to its value. Crediting its net realisable value to the process achieves the correct main-product cost with one line — proportionality, not unfairness.